Stage 4 — Check-your-understanding Answers

Q1. The pseudo-true order

The pseudo-true order alpha* is the fractional order maximizing the expected log-likelihood (equivalently, minimizing Kullback–Leibler divergence) when the assumed Caputo SEIR family is fitted to the true data-generating mechanism. Fitting expected weekly counts removes sampling fluctuations and isolates the limiting effect of model misspecification: with increasingly informative data, the fitted order approaches alpha*, not necessarily a true mechanistic order.

Table 6: genuine Caputo memory (T4, alpha = 0.8) gives alpha* = 0.800, while memoryless heterogeneous recovery (T7) gives alpha* = 0.810. Their absolute difference is 0.010. Thus two distinct mechanisms can yield essentially the same fractional-order estimate even with unlimited data: alpha below one alone is not proof of genuine memory.

Q2. The mixture of exponentials, by hand

For equally sized recovery classes with rates 2 and 2/3 per week, phi(t) = (1/2) exp(-2t) + (1/2) exp(-2t/3).

Mean infectious period: integral from 0 to infinity of phi(t) dt = (1/2)(1/2) + (1/2)(3/2) = 1 week. The equal-mean single-rate exponential is exp(-t).

At t = 1: phi(1) = 0.5(0.135335) + 0.5(0.513417) = 0.324376; exp(-1) = 0.367879. At t = 3: phi(3) = 0.5(0.002479) + 0.5(0.135335) = 0.068907; exp(-3) = 0.049787. The mixture survival at week 3 is 0.068907/0.049787 = 1.384 times larger.

The slow class dominates the late tail: its contribution at t = 3 is 0.067668, compared with 0.001239 from the fast class. This prolonged infectiousness can mimic fractional recovery, explaining T7's pseudo-true alpha* = 0.810. Proposition 11(i) proves the exact continuous-mixture representation for the AHM survival function; the two-class calculation is an illustrative approximation, not an exact power-law tail.

Q3. Why the test needs calibration

A size of 12% means that 12 of every 100 samples generated by the ordinary SEIR truth would falsely reject alpha = 1, even though the intended false-positive rate is 5%. This is an excess of 7 percentage points (2.4 times the target).

The likelihood-ratio statistic is Lambda = 2[ell_max - ell_p(1)]. The nominal boundary reference is a 50:50 mixture of a point mass at zero and chi-square with one degree of freedom, giving critical value 2.71. For negative-binomial dispersion k = 10, simulate 200 samples from ordinary SEIR (T0), refit both models, calculate Lambda for each, and use the empirical 95th percentile: 4.43. Figure 11(a) shows why 2.71 is too low for this setting.

The bootstrap calibration is conditional on ordinary SEIR being true. Erlang-stage and reporting-delay truths are misspecified relative to that null; their likelihood-ratio distributions and pseudo-true parameters need not follow the T0 calibration (the concern highlighted by misspecified-model likelihood theory, including Vuong). With 200 replicates, the approximate 95% margin for a measured 12% rejection rate is 1.96 sqrt(0.12*0.88/200) = 0.045, or +/-4.5 percentage points; at 5%, it is +/-3.0 points.

Q4. Before or after the peak?

Declining transmission: 41.0/53.9 x 100 = 76.07% of the log-likelihood gain comes from the pre-peak period. Heterogeneous recovery: 4.6/14.3 x 100 = 32.17% comes before the peak, so (14.3 - 4.6)/14.3 x 100 = 67.83% comes afterward (the paper rounds the latter to 68%, using 9.8/14.3).

For declining transmission, the fractional order mimics slowing growth during the rising phase: Proposition 6 shows that fractional onset depends on t^alpha. For heterogeneous recovery, it mimics the slower post-peak decline created by persistent slow recoverers. Therefore a series stopping four weeks after its peak should accentuate the first imitation and weaken the second.

Table 7 confirms this: T2 declining transmission changes from alpha* = 0.763 (full) to alpha* <= 0.40 (truncated, at the search-grid boundary), whereas T7 heterogeneous recovery changes from 0.810 to 0.936 (about 0.94). A short tail hides much of the recovery-heterogeneity signature.

Q5. When cumulative cases fall

Remark 4 writes S(t) = S(0) - [1/Gamma(alpha)] integral from 0 to t of (t-s)^(alpha-1) F(s) ds, with F(s) = beta^alpha S(s)I(s)/N >= 0. For alpha = 0.9, the kernel is (t-s)^(-0.1). At lag 1 it is 1^(-0.1) = 1; at lag 10 it is 10^(-0.1) = 0.794328, approximately 0.79. Thus a past infection's contribution to the integral shrinks as its age increases.

If new infections become sufficiently small, the weighted integral can decrease despite F(s) >= 0, so S(t) rises and the putative cumulative quantity I(t)+R(t) = N-S(t) falls. This is a mathematical memory effect, not a biological return to susceptibility. At alpha = 1 the kernel is 1 and S(t) = S(0) - integral F(s) ds; because F >= 0, S cannot rise and cumulative infections cannot fall.

In the paper's alpha = 0.9 example, S reaches its minimum in week 29 and then rises by about 12.5% of N by week 150; the first negative weekly increment of I+R occurs in week 33. Negative predicted weekly cases are invalid as count means. Accordingly, the fitting procedure assigns such weeks a small positive mean floor of 10^(-12), imposing a severe likelihood penalty. This means a Caputo fit to incidence must be checked for biologically impossible negative increments, even if all compartment values remain nonnegative.